

INFO:tensorflow:Reduce to /job:localhost/replica:0/task:0/device:CPU:0 then broadcast to ('/job:localhost/replica:0/task:0/device:CPU:0').
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[CHECKPOINT] Latest saved after epoch 1
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 1: val_loss = 0.0037864617
[BEST] File size: 12.41 MB
3972/3972 ————— 335s 83ms/step - loss: 0.0086 - val_loss: 0.0038
Epoch 2/50
3972/3972 ————— 0s 81ms/step - loss: 0.0039
[CHECKPOINT] Latest saved after epoch 2
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 2: val_loss = 0.0028494573
[BEST] File size: 12.41 MB
3972/3972 ————— 340s 85ms/step - loss: 0.0038 - val_loss: 0.0028
Epoch 3/50
3972/3972 ————— 0s 80ms/step - loss: 0.0026
[CHECKPOINT] Latest saved after epoch 3
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 3: val_loss = 0.0016845805
[BEST] File size: 12.41 MB
3972/3972 ————— 336s 84ms/step - loss: 0.0023 - val_loss: 0.0017
Epoch 4/50
3972/3972 ————— 0s 81ms/step - loss: 0.0018
[CHECKPOINT] Latest saved after epoch 4
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 4: val_loss = 0.0013302859
[BEST] File size: 12.41 MB
3972/3972 ————— 341s 85ms/step - loss: 0.0017 - val_loss: 0.0013
Epoch 5/50
3970/3972 ————— 0s 80ms/step - loss: 0.0013
[CHECKPOINT] Latest saved after epoch 5
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 5: val_loss = 0.0009965121
[BEST] File size: 12.41 MB
3972/3972 ————— 339s 85ms/step - loss: 0.0012 - val_loss: 9.9651e-04
Epoch 6/50
3970/3972 ————— 0s 81ms/step - loss: 0.0011
[CHECKPOINT] Latest saved after epoch 6
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 6: val_loss = 0.0009405880
[BEST] File size: 12.41 MB
3972/3972 ————— 340s 85ms/step - loss: 0.0011 - val_loss: 9.4059e-04
Epoch 7/50
3972/3972 ————— 0s 77ms/step - loss: 9.3780e-04
[CHECKPOINT] Latest saved after epoch 7
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 7: val_loss = 0.0008805282
[BEST] File size: 12.41 MB
3972/3972 ————— 326s 82ms/step - loss: 9.2419e-04 - val_loss: 8.8053e-04
Epoch 8/50
3971/3972 ————— 0s 77ms/step - loss: 9.3583e-04

[CHECKPOINT] Latest saved after epoch 8
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 325s 81ms/step - loss: 0.0011 - val_loss: 8.9057e-04
Epoch 9/50
3972/3972 ————— 0s 79ms/step - loss: 8.7253e-04
[CHECKPOINT] Latest saved after epoch 9
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 9: val_loss = 0.0007288713
[BEST] File size: 12.41 MB
3972/3972 ————— 334s 84ms/step - loss: 8.3150e-04 - val_loss: 7.2887e-04
Epoch 10/50
3971/3972 ————— 0s 82ms/step - loss: 8.9292e-04
[CHECKPOINT] Latest saved after epoch 10
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 344s 86ms/step - loss: 9.3499e-04 - val_loss: 7.8581e-04
Epoch 11/50
3972/3972 ————— 0s 79ms/step - loss: 7.8651e-04
[CHECKPOINT] Latest saved after epoch 11
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 11: val_loss = 0.0007068779
[BEST] File size: 12.41 MB
3972/3972 ————— 334s 83ms/step - loss: 7.7713e-04 - val_loss: 7.0688e-04
Epoch 12/50
3970/3972 ————— 0s 77ms/step - loss: 9.2080e-04
[CHECKPOINT] Latest saved after epoch 12
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 12: val_loss = 0.0006529019
[BEST] File size: 12.41 MB
3972/3972 ————— 324s 81ms/step - loss: 8.2953e-04 - val_loss: 6.5290e-04
Epoch 13/50
3971/3972 ————— 0s 80ms/step - loss: 8.9918e-04
[CHECKPOINT] Latest saved after epoch 13
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 336s 84ms/step - loss: 8.1961e-04 - val_loss: 7.3903e-04
Epoch 14/50
3970/3972 ————— 0s 79ms/step - loss: 0.0015
[CHECKPOINT] Latest saved after epoch 14
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 335s 84ms/step - loss: 0.0011 - val_loss: 6.6890e-04
Epoch 15/50
3970/3972 ————— 0s 76ms/step - loss: 7.5557e-04
[CHECKPOINT] Latest saved after epoch 15
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 15: val_loss = 0.0006276334
[BEST] File size: 12.41 MB
3972/3972 ————— 322s 80ms/step - loss: 7.3984e-04 - val_loss: 6.2763e-04
Epoch 16/50
3972/3972 ————— 0s 76ms/step - loss: 7.1754e-04
[CHECKPOINT] Latest saved after epoch 16
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 16: val_loss = 0.0006072237
[BEST] File size: 12.41 MB
3972/3972 ————— 322s 81ms/step - loss: 6.9832e-04 - val_loss: 6.0722e-04
Epoch 17/50
3970/3972 ————— 0s 78ms/step - loss: 9.3081e-04

[CHECKPOINT] Latest saved after epoch 17

[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 17: val_loss = 0.0005859251

[BEST] File size: 12.41 MB

3972/3972 ————— 330s 83ms/step - loss: 8.1294e-04 - val_loss: 5.8593e-04

Epoch 18/50

3970/3972 ————— 0s 84ms/step - loss: 7.3226e-04

[CHECKPOINT] Latest saved after epoch 18

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 354s 88ms/step - loss: 7.3187e-04 - val_loss: 7.0911e-04

Epoch 19/50

3972/3972 ————— 0s 87ms/step - loss: 6.2712e-04

[CHECKPOINT] Latest saved after epoch 19

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 365s 91ms/step - loss: 7.1411e-04 - val_loss: 6.7768e-04

Epoch 20/50

3970/3972 ————— 0s 83ms/step - loss: 6.4082e-04

[CHECKPOINT] Latest saved after epoch 20

[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 20: val_loss = 0.0005412861

[BEST] File size: 12.41 MB

3972/3972 ————— 351s 88ms/step - loss: 6.1675e-04 - val_loss: 5.4129e-04

Epoch 21/50

3972/3972 ————— 0s 85ms/step - loss: 9.3115e-04

[CHECKPOINT] Latest saved after epoch 21

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 357s 89ms/step - loss: 7.6814e-04 - val_loss: 5.7751e-04

Epoch 22/50

3971/3972 ————— 0s 82ms/step - loss: 8.3440e-04

[CHECKPOINT] Latest saved after epoch 22

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 347s 87ms/step - loss: 6.8934e-04 - val_loss: 5.6263e-04

Epoch 23/50

3970/3972 ————— 0s 85ms/step - loss: 6.4746e-04

[CHECKPOINT] Latest saved after epoch 23

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 359s 90ms/step - loss: 6.7405e-04 - val_loss: 6.1046e-04

Epoch 24/50

3970/3972 ————— 0s 86ms/step - loss: 6.0124e-04

[CHECKPOINT] Latest saved after epoch 24

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 365s 91ms/step - loss: 7.6467e-04 - val_loss: 0.0011

Epoch 25/50

3971/3972 ————— 0s 89ms/step - loss: 6.2312e-04

[CHECKPOINT] Latest saved after epoch 26

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 372s 93ms/step - loss: 6.9951e-04 - val_loss: 5.5771e-04

Epoch 27/50

3971/3972 ————— 0s 85ms/step - loss: 5.6515e-04

[CHECKPOINT] Latest saved after epoch 27

[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 27: val_loss = 0.0005167282

[BEST] File size: 12.41 MB

3972/3972 ————— 361s 90ms/step - loss: 5.5606e-04 - val_loss: 5.1673e-04

Epoch 28/50

3971/3972 ————— 0s 91ms/step - loss: 5.4467e-04

[CHECKPOINT] Latest saved after epoch 28
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 383s 96ms/step - loss: 7.1603e-04 - val_loss: 8.8743e-04
Epoch 29/50
3972/3972 ————— 0s 89ms/step - loss: 6.9418e-04
[CHECKPOINT] Latest saved after epoch 29
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 373s 93ms/step - loss: 6.2518e-04 - val_loss: 5.5096e-04
Epoch 30/50
3970/3972 ————— 0s 91ms/step - loss: 5.8017e-04
[CHECKPOINT] Latest saved after epoch 30
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 381s 95ms/step - loss: 6.8770e-04 - val_loss: 6.3808e-04
Epoch 31/50
3972/3972 ————— 0s 92ms/step - loss: 5.9103e-04
[CHECKPOINT] Latest saved after epoch 31
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 31: val_loss = 0.0005157573
[BEST] File size: 12.41 MB
3972/3972 ————— 386s 97ms/step - loss: 5.6525e-04 - val_loss: 5.1576e-04
Epoch 32/50
3972/3972 ————— 0s 90ms/step - loss: 5.2972e-04
[CHECKPOINT] Latest saved after epoch 32
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 32: val_loss = 0.0005148580
[BEST] File size: 12.41 MB
3972/3972 ————— 380s 95ms/step - loss: 5.2495e-04 - val_loss: 5.1486e-04
Epoch 33/50
3972/3972 ————— 0s 87ms/step - loss: 5.7666e-04
[CHECKPOINT] Latest saved after epoch 33
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 366s 91ms/step - loss: 7.9957e-04 - val_loss: 7.4355e-04
Epoch 34/50
3971/3972 ————— 0s 88ms/step - loss: 6.7482e-04
[CHECKPOINT] Latest saved after epoch 34
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 371s 93ms/step - loss: 6.2591e-04 - val_loss: 6.0278e-04
Epoch 35/50
3970/3972 ————— 0s 89ms/step - loss: 7.3068e-04
[CHECKPOINT] Latest saved after epoch 35
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 376s 94ms/step - loss: 7.6333e-04 - val_loss: 5.9133e-04
Epoch 36/50
3972/3972 ————— 0s 92ms/step - loss: 8.0367e-04
[CHECKPOINT] Latest saved after epoch 36
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 386s 96ms/step - loss: 8.2344e-04 - val_loss: 6.0149e-04
Epoch 37/50
3971/3972 ————— 0s 89ms/step - loss: 6.9201e-04
[CHECKPOINT] Latest saved after epoch 37
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 376s 94ms/step - loss: 9.8181e-04 - val_loss: 8.4179e-04
Epoch 38/50
3970/3972 ————— 0s 89ms/step - loss: 7.3383e-04
[CHECKPOINT] Latest saved after epoch 38
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 376s 94ms/step - loss: 6.6974e-04 - val_loss: 5.7284e-04

Epoch 39/50
3971/3972 ————— 0s 89ms/step - loss: 7.1738e-04
[CHECKPOINT] Latest saved after epoch 39
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 374s 93ms/step - loss: 7.4862e-04 - val_loss: 6.3565e-04
Epoch 40/50
3972/3972 ————— 0s 88ms/step - loss: 5.8574e-04
[CHECKPOINT] Latest saved after epoch 40
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 372s 93ms/step - loss: 5.7313e-04 - val_loss: 5.2009e-04
Epoch 41/50
3970/3972 ————— 0s 89ms/step - loss: 5.4448e-04
[CHECKPOINT] Latest saved after epoch 41
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 373s 93ms/step - loss: 5.3897e-04 - val_loss: 5.6095e-04
Epoch 42/50
3972/3972 ————— 0s 87ms/step - loss: 6.2237e-04
[CHECKPOINT] Latest saved after epoch 42
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 366s 91ms/step - loss: 6.3179e-04 - val_loss: 5.3942e-04
Epoch 43/50
3970/3972 ————— 0s 87ms/step - loss: 5.2824e-04
[CHECKPOINT] Latest saved after epoch 43
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 43: val_loss = 0.0005004936
[BEST] File size: 12.41 MB
3972/3972 ————— 366s 91ms/step - loss: 5.2106e-04 - val_loss: 5.0049e-04
Epoch 44/50
3972/3972 ————— 0s 86ms/step - loss: 5.6177e-04
[CHECKPOINT] Latest saved after epoch 44
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 44: val_loss = 0.0004901169
[BEST] File size: 12.41 MB
3972/3972 ————— 365s 91ms/step - loss: 5.7325e-04 - val_loss: 4.9012e-04
Epoch 45/50
3970/3972 ————— 0s 87ms/step - loss: 5.1601e-04
[CHECKPOINT] Latest saved after epoch 45
[CHECKPOINT] File size: 12.41 MB
3972/3972 ————— 366s 91ms/step - loss: 5.0923e-04 - val_loss: 4.9312e-04
Epoch 46/50
3970/3972 ————— 0s 92ms/step - loss: 6.5135e-04
[CHECKPOINT] Latest saved after epoch 46
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 46: val_loss = 0.0004656877
[BEST] File size: 12.41 MB
3972/3972 ————— 385s 96ms/step - loss: 5.5522e-04 - val_loss: 4.6569e-04
Epoch 47/50
3971/3972 ————— 0s 89ms/step - loss: 5.1288e-04
[CHECKPOINT] Latest saved after epoch 47
[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 47: val_loss = 0.0004601577
[BEST] File size: 12.41 MB
3972/3972 ————— 374s 93ms/step - loss: 5.0343e-04 - val_loss: 4.6016e-04
Epoch 48/50
3971/3972 ————— 0s 88ms/step - loss: 5.0769e-04

[CHECKPOINT] Latest saved after epoch 48

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 372s 93ms/step - loss: 5.8230e-04 - val_loss: 4.9303e-04

Epoch 49/50

3972/3972 ————— 0s 88ms/step - loss: 4.9345e-04

[CHECKPOINT] Latest saved after epoch 49

[CHECKPOINT] File size: 12.41 MB

[BEST] Epoch 49: val_loss = 0.0004580515

[BEST] File size: 12.41 MB

3972/3972 ————— 371s 93ms/step - loss: 4.8511e-04 - val_loss: 4.5805e-04

Epoch 50/50

3970/3972 ————— 0s 88ms/step - loss: 4.7972e-04

[CHECKPOINT] Latest saved after epoch 50

[CHECKPOINT] File size: 12.41 MB

3972/3972 ————— 371s 93ms/step - loss: 5.4033e-04 - val_loss: 5.1840e-04

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FINAL MODEL SAVE

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VERIFYING FINAL ARTIFACTS

✓ /kaggle/working/AE_280K/models/AE_280K_final.keras

Size: 12.41 MB

✓ /kaggle/working/AE_280K/models/AE_280K_encoder.keras

Size: 1.50 MB

✓ /kaggle/working/AE_280K/logs/history.json

Size: 0.00 MB

✓ /kaggle/working/AE_280K/logs/training_state.json

Size: 0.00 MB

✓ /kaggle/working/AE_280K/models/AE_280K_best.keras

Size: 12.41 MB

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TRAINING COMPLETE

Completed epochs: 50

Best epoch: 49

Best val_loss: 0.00045805147965438664

Final optimizer iterations: 198650
